

Flatcar Updates & Releases

Source section: `updates-releases/` (consolidates `nebraska/` and `setup/releases/` from the main branch into one section in the refactored branch).

Release Channels (`updates-releases/releases/`)

Channel	Cadence	Purpose
Alpha	Frequent	Latest features, cutting edge
Beta	~2 months	Release candidates; no known issues
Stable	~2 months	Production-ready
LTS	12 months	18-month security support (commercial)

Switching channels via Butane:

```
variant: flatcar
version: 1.0.0
storage:
  files:
    - path: /etc/flatcar/update.conf
      overwrite: true
      mode: 0644
      contents:
        inline: |
          GROUP=beta
          SERVER=https://public.update.flatcar-linux.net/v1/update/
```

Update Flow

1. `update-engine.service` downloads update to the **inactive** USR partition (background)
2. Reboot manager coordinates when the node reboots
3. GRUB boots into the updated partition
4. On boot failure, automatic rollback to the previous partition

update-engine status values:

IDLE → CHECKING_FOR_UPDATE → UPDATE_AVAILABLE → DOWNLOADING → VERIFYING → FINALIZING → UPDATED_NEED_REBOOT

Force an immediate update check:

```
update_engine_client -update
```

Reboot Strategies (updates - releases / releases / update - strategies .md)

Strategy	Behavior
reboot (default)	Reboot 5 minutes after update is applied
etcd-lock	Acquire distributed etcd lock first
off	Never reboot automatically

Reboot manager options:

- `locksmithd` — default, included in image
- **FLUO** ([flatcar-linux-update-operator](#)) — recommended for Kubernetes; understands node draining
- **Kured** — supported from Flatcar >3067.0.0

Reboot window example (locksmithd):

```
storage:
  files:
    - path: /etc/flatcar/update.conf
      contents:
        inline: |
          REBOOT_STRATEGY=reboot
          LOCKSMITHD_REBOOT_WINDOW_START=02:00
          LOCKSMITHD_REBOOT_WINDOW_LENGTH=1h
```

update.conf (updates - releases / releases / update - conf .md)

Key fields in `/etc/flatcar/update.conf` :

Field	Purpose
SERVER	Update server URL
GROUP	Channel name or Nebraska group UUID
REBOOT_STRATEGY	Reboot coordination strategy
MACHINE_ALIAS	Display name in Nebraska dashboard

Verifying Images (updates - releases / releases / verify - images .md)

Flatcar images are GPG-signed. Verify before use:

```
gpg --verify flatcar_production_image.vmlinuz.sig
```

Nebraska Update Manager (updates - releases / nebraska /)

Nebraska is a self-hosted Omaha-protocol update server for Flatcar. It gives operators control over which nodes update, when, and at what rate.

What Nebraska provides:

- Dashboard to monitor instance update status
- Instance groups with configurable channels (alpha/beta/stable or custom UUIDs)
- Rollout rate control and time windows
- Machine alias display
- OEM extension payload serving (required since Flatcar 3760.0.0)

Pointing new machines at Nebraska (Butane):

```
variant: flatcar
version: 1.0.0
storage:
  files:
    - path: /etc/flatcar/update.conf
      overwrite: true
      mode: 0644
      contents:
        inline: |
          GROUP=stable
          SERVER=http://your.nebraska.host:8000/v1/update/
```

Custom Nebraska group:

```
GROUP=ab51a000-02dc-4fc7-a6b0-c42881c89856
SERVER=http://your.nebraska.host:8000/v1/update/
```

Existing machines:

```
# Edit /etc/flatcar/update.conf
echo "SERVER=http://your.nebraska.host:8000/v1/update/" >>
/etc/flatcar/update.conf
update_engine_client -update
```

Nebraska is open source: <https://github.com/flatcar/nebraska>

Default sample port: 8000, plain HTTP — configure TLS for production.

OEM payloads: Self-hosted Nebraska must serve OEM sysexr payloads (introduced in Flatcar 3760.0.0). Keep Nebraska up to date.